



2025 年 10 月 CSCA 考试数学试卷

选择题(Multiple Choice Questions)

每题有 4 个选项，其中只有 1 个选项正确。

(Each question has 4 options, of which only 1 option is correct.)

1. 下列说法正确的是? (Which of the following statements is correct?)

- A. $-2 \in \mathbb{N}$
- B. $\pi \in \mathbb{Q}$
- C. $\frac{3}{2} \in \mathbb{Z}$
- D. $\sqrt{3} \in \mathbb{R}$

2. 已知集合 $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, 则下列运算正确的是? (If sets $A = \{1, 2, 3, 4\}$,

$B = \{3, 4, 5, 6\}$, then which of the following operations is correct?)

- A. $A \cup B = 3, 4$
- B. $A \cap B = 5, 6$
- C. $A \cap B = 1, 2, 5, 6$
- D. $A \cup B = 1, 2, 3, 4, 5, 6$

3. 函数 $f(x) = \ln x + \sqrt{1-x}$ 的定义域是? (The domain of the function $f(x) = \ln x + \sqrt{1-x}$ is?)

- A. $\{x \mid 0 < x \leq 1\}$
- B. $x \mid x > 0$
- C. $x \mid x \leq 1$
- D. $x \mid x \geq 1$

4. 若点 $M(a+3, a+1)$ 在 x 轴上, 则该点的坐标是? (If point $M(a+3, a+1)$ is on the x -axis, then the coordinates of the point are?)

- A. $(0, -4)$
- B. $(0, -2)$
- C. $(2, 0)$
- D. $(4, 0)$



5. 已知角 $\alpha = \frac{5\pi}{6}$, 下列结论正确的是? (Given that angle $\alpha = \frac{5\pi}{6}$, which of the following conclusions is correct?)
- A. $\sin\alpha = \frac{1}{2}$
- B. $\sin\alpha = -\frac{1}{2}$
- C. $\sin\alpha = \frac{\sqrt{3}}{2}$
- D. $\sin\alpha = -\frac{\sqrt{3}}{2}$
6. 下列哪个函数是偶函数? (Which of the following functions is an even function?)
- A. $f(x) = 2x^2 + \cos x$
- B. $f(x) = x + 3x^2$
- C. $f(x) = \sqrt{x}$
- D. $f(x) = x^2 - \sin x$
7. 下列哪个点位于第三象限? (Which of the following points is located in the third quadrant?)
- A. (1, -2)
- B. (-1, -2)
- C. (1, 2)
- D. (-1, 2)
8. 已知函数 $y = 1 - x^3$, 其反函数是? (Given the function $y = 1 - x^3$, its inverse function is?)
- A. $y = \sqrt[3]{1-x}$
- B. $y = \sqrt[3]{x-1}$
- C. $y = \sqrt[3]{1+x}$
- D. $y = \sqrt{1-x}$
9. 下列哪个函数在 $(0, +\infty)$ 上是减函数? (Which of the following functions is decreasing



on $(0, +\infty)$?)

- A. $y = 3x$
- B. $y = x^2$
- C. $y = x^3$
- D. $y = \frac{1}{x}$

10. 有理不等式 $\frac{2x-4}{x-1} \leq 0$ 的解集是? (The solution set of the rational inequality $\frac{2x-4}{x-1} \leq 0$

is?)

- A. $x \mid 1 < x \leq 2$
- B. $x \mid 1 \leq x < 2$
- C. $x \mid 1 \leq x \leq 2$
- D. $x \mid x < 1$ 或 $x > 2$ ($x \mid x < 1$ or $x > 2$)

11. 已知 $a = \sqrt{2} + 1$, $b = \sqrt{2} - 1$, 则 a 和 b 的几何平均数是? (Given $a = \sqrt{2} + 1$ and $b = \sqrt{2} - 1$, the geometric mean of a and b is?)

- A. ± 1
- B. 1
- C. $\sqrt{2}$
- D. -1

12. 若 a_n 是首项 $a_1 = 2$ 、公差 $d = 3$ 的等差数列, 则 $a_{100} = ?$ (If a_n is an arithmetic sequence with first term $a_1 = 2$ and common difference $d = 3$, then $a_{100} = ?$)

- A. 299
- B. 302
- C. 298
- D. 305

13. 经过点 $A(-1,1)$ 和 $B(2,3)$ 的直线的斜率是? (The slope of the line passing through points $A(-1,1)$ and $B(2,3)$ is?)

- A. 3



B. $\frac{2}{3}$

C. $\frac{3}{2}$

D. $-\frac{2}{3}$

14. 已知角 α 的终边经过点 $M(2,4)$, 则 $\tan\alpha = ?$ (Given that the terminal side of angle α passes through point $M(2,4)$, then $\tan\alpha = ?$)

A. 2

B. -2

C. $\frac{1}{2}$

D. $-\frac{1}{2}$

15. 若数列 a_n 是首项 $a_1 = 1$ 、 $a_2 = 3$ 的等差数列, 则 $a_5 = ?$ (If the sequence a_n is an arithmetic sequence with $a_1 = 1$ and $a_2 = 3$, then $a_5 = ?$)

A. 9

B. 11

C. 5

D. 7

16. 已知点 $A(-1,2)$ 和 $B(3,-1)$, 则这两点间的距离 $|AB| = ?$ (Given points $A(-1,2)$ and $B(3,-1)$, the distance between these two points $|AB| = ?$)

A. $\sqrt{5}$

B. 25

C. 5

D. 4

17. 经过点 $M(-2,3)$ 且斜率 $k = 2$ 的直线方程是? (The equation of the line passing



through point $M(-2,3)$ with slope $k = 2$ is?)

- A. $y = x + 7$
- B. $y = 2x - 1$
- C. $y = 2x + 7$
- D. $y = x + 5$

18. 若一个圆经过点 $A(2,2)$ 且圆心为 $(2,0)$, 则该圆的方程是? (If a circle passes through point $A(2,2)$ and has center $(2,0)$, then the equation of the circle is?)

- A. $(x - 2)^2 + y^2 = 2$
- B. $x^2 + (y - 2)^2 = 2$
- C. $(x - 2)^2 + (y - 2)^2 = 4$
- D. $(x - 2)^2 + y^2 = 4$

19. 若 $\tan \alpha = 2$, 则 $\frac{3\sin \alpha + 2\cos \alpha}{4\sin \alpha - 3\cos \alpha} = ?$ (If $\tan \alpha = 2$, then $\frac{3\sin \alpha + 2\cos \alpha}{4\sin \alpha - 3\cos \alpha} = ?$)

- A. $\frac{8}{5}$
- B. $-\frac{8}{5}$
- C. $\frac{5}{3}$
- D. $-\frac{5}{3}$

20. 对于数列 $a_n : \frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \dots$, 其通项公式是? (For a sequence $a_n : \frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \dots$, the general term is $a_n = ?$)

- A. $(-1)^{n-1} \cdot \frac{1}{2^n}$
- B. $\frac{1}{2^n}$
- C. $(-1)^{n-1} \cdot \frac{1}{2^{n-1}}$
- D. $\frac{1}{2^{n-1}}$

21. 若 a, b, c 是有理数且 $a > b > c$, 则下列表达式正确的是? (If a, b, c are rational numbers and $a > b > c$, then which of the following expressions is correct?)

- A. $\frac{a}{c} < \frac{b}{c}$
- B. $a + b > b + c$
- C. $a - b > b - c$



D. $ab > bc$

22. 关于幂函数 $y = x^3$ 和 $y = x^{\frac{1}{3}}$, 下列结论正确的是? (Regarding the power functions

$y = x^3$ and $y = x^{\frac{1}{3}}$, which of the following conclusions is correct?)

A. 两个函数的图像关于 x 轴对称(The graphs of the two functions are symmetric about the x -axis)

B. 两个函数的图像关于直线 $y = x$ 对称(The graphs of the two functions are symmetric about the line $y = x$)

C. 两个函数的图像关于 y 轴对称(The graphs of the two functions are symmetric about the y -axis)

D. 两个函数的图像关于原点对称(The graphs of the two functions are symmetric about the origin)

23. 若 $\cos\alpha = \frac{1}{3}$, 则 $\cos 2\alpha = ?$ (If $\cos\alpha = \frac{1}{3}$, then $\cos 2\alpha = ?$)

A. $\frac{4\sqrt{2}}{9}$

B. $-\frac{7}{9}$

C. $-\frac{4\sqrt{2}}{9}$

D. $\frac{7}{9}$

24. 直线 $3x + 2y + 4 = 0$ 和 $x - y + 3 = 0$ 的交点坐标是? (The coordinates of the intersection point of lines $3x + 2y + 4 = 0$ and $x - y + 3 = 0$ are?)

A. (1, -2)

B. (2, 5)

C. (-2, 1)

D. (5, 2)

25. 双曲线 $\frac{x^2}{7} - \frac{y^2}{9} = 1$ 的焦点坐标是? (The foci of the hyperbola $\frac{x^2}{7} - \frac{y^2}{9} = 1$ are?)



A. (0, 4)

B. (4, 0)

C. (0, ± 4)

D. (± 4 , 0)

26. 下列不等式正确的是? (Which of the following inequalities is correct?)

A. $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$

B. $2.1^{-2} > 1.2^{-2}$

C. $0.75^{-0.2} > 0.75^{-0.4}$

D. $3^{2.25} > 3^3$

27. $\frac{\log_4 27}{\log_2 3} = ?$

A. $\frac{3}{2}$

B. $-\frac{3}{2}$

C. $\frac{2}{3}$

D. $-\frac{2}{3}$

28. 关于余弦函数 $y = \cos x$, 下列结论正确的是? (Regarding the cosine function $y = \cos x$, which of the following conclusions is correct?)

A. 该函数是周期函数(The function is a periodic function)

B. 该函数是减函数(The function is a decreasing function)

C. 该函数是增函数(The function is an increasing function)

D. 该函数是奇函数(The function is an odd function)

29. 顶点在原点、焦点为 $F\left(\frac{3}{2}, 0\right)$ 的抛物线的标准方程是? (The standard equation of the parabola with vertex at the origin and focus $F\left(\frac{3}{2}, 0\right)$ is?)

A. $y^2 = \frac{3}{2}x$

B. $y^2 = 3x$



- C. $y^2 = 6x$
D. $y^2 = -6x$

30. 点 $P(-1,2)$ 到点 $Q(3,1)$ 的距离是? (The distance from point $P(-1,2)$ to point $Q(3,1)$ is?)

- A. $\sqrt{5}$
B. $\sqrt{13}$
C. $\sqrt{17}$
D. 5

31. 已知直线 $l_1: ax + 2y - 1 = 0$ 和 $l_2: x + (a + 1)y + 4 = 0$ 平行, 则 $a = ?$ (Given that lines $l_1: ax + 2y - 1 = 0$ and $l_2: x + (a + 1)y + 4 = 0$ are parallel, then $a = ?$)

- A. 1
B. -2 或 1(-2 or 1)
C. -2
D. 2 或 -1(2 or -1)

32. 若圆的圆心在 $(-3,2)$ 且半径为 4, 则该圆的方程是? (If the center of a circle is at $(-3,2)$ and the radius is 4, then the equation of the circle is?)

- A. $(x - 3)^2 + (y + 2)^2 = 4$
B. $(x - 3)^2 + (y + 2)^2 = 16$
C. $(x + 3)^2 + (y - 2)^2 = 4$
D. $(x + 3)^2 + (y - 2)^2 = 16$

33. 已知 $\cos \alpha = -\frac{4}{5}$ 且 $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, 则 $\sin\left(\alpha + \frac{\pi}{4}\right) = ?$ (Given that $\cos \alpha = -\frac{4}{5}$ and $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, then $\sin\left(\alpha + \frac{\pi}{4}\right) = ?$)

- A. $\frac{7\sqrt{2}}{10}$
B. $-\frac{\sqrt{2}}{10}$
C. $\frac{\sqrt{2}}{10}$



D. $\frac{7\sqrt{2}}{10}$

34. 下列哪组中的两个函数是同一个函数? (In which of the following pairs are the two functions identical?)

A. $y = \ln x^2$ 和 $y = 2\ln x$ ($y = \ln x^2$ and $y = 2\ln x$)

B. $y = |x - 1|$ 和 $y = \sqrt{(x - 1)^2}$ ($y = |x - 1|$ and $y = \sqrt{(x - 1)^2}$)

C. $y = \frac{x^4 - 1}{x^2 - 1}$ 和 $y = x + 1$ ($y = \frac{x^4 - 1}{x^2 - 1}$ and $y = x + 1$)

D. $y = \ln[x(x - 1)]$ 和 $y = \ln x + \ln(x - 1)$ ($y = \ln[x(x - 1)]$ and $y = \ln x + \ln(x - 1)$)

35. 已知 $\sin \alpha = \frac{3}{5}$ 且 $\alpha \in (0, \frac{\pi}{2})$, 则 $\frac{2\sin 2\alpha}{1 + \cos 2\alpha} - \tan \alpha = ?$ (Given that $\sin \alpha = \frac{3}{5}$ and $\alpha \in$

$(0, \frac{\pi}{2})$, then $\frac{2\sin 2\alpha}{1 + \cos 2\alpha} - \tan \alpha = ?$)

A. $\frac{3}{5}$

B. $\frac{4}{3}$

C. $\frac{3}{4}$

D. $\frac{4}{5}$

36. 下列表达式正确的是? (Which of the following expressions is correct?)

A. $\sin(\alpha + 4\pi) = \cos \alpha$

B. $\sin(\alpha + 3\pi) = \sin \alpha$

C. $\cos(\alpha + \frac{3\pi}{2}) = \sin \alpha$

D. $\cos(\alpha + \frac{\pi}{2}) = \sin \alpha$

37. 关于正切函数 $y = \tan x$, 下列说法不正确的是? (Regarding the tangent function $y = \tan x$, which of the following statements is incorrect?)

A. 值域为 $\mathbb{R}$ (The range is $\mathbb{R}$)

B. 是奇函数(It is an odd function)



- C. 是最小正周期为 π 的周期函数 (It is a periodic function with minimum period π)
- D. 定义域为 $\{x \mid x \neq k\pi + \frac{\pi}{2}, k \in \mathbb{Z}\}$ (The domain is $\{x \mid x \neq k\pi + \frac{\pi}{2}, k \in \mathbb{Z}\}$)
38. 已知 $\cos\alpha = -\frac{1}{2}$ 且 $\alpha \in (\frac{\pi}{2}, \pi)$, 则 $\sin\frac{\alpha}{2} = ?$ (Given that $\cos\alpha = -\frac{1}{2}$ and $\alpha \in (\frac{\pi}{2}, \pi)$, the value of $\sin\frac{\alpha}{2}$ is?)
- A. $-\frac{1}{2}$
- B. $-\frac{\sqrt{3}}{2}$
- C. $\frac{1}{2}$
- D. $\frac{\sqrt{3}}{2}$
39. 对于数列 a_n , 已知 $a_1 = 1$, $a_2 = 3$, 且 $a_{n+2} = a_{n+1} + a_n$ ($n = 1, 2, \dots$), 则 $a_5 = ?$
- (For the sequence a_n with $a_1 = 1$, $a_2 = 3$, and $a_{n+2} = a_{n+1} + a_n$ ($n = 1, 2, \dots$), then $a_5 = ?$)
- A. 7
- B. -7
- C. 11
- D. -11
40. 抛物线 $y = -x^2$ 的准线方程是? (The directrix equation of the parabola $y = -x^2$ is?)
- A. $y = \frac{1}{4}$
- B. $y = -\frac{1}{2}$
- C. $y = -\frac{1}{4}$
- D. $y = \frac{1}{2}$
41. 已知 $a = \ln 3$, $b = \log_{0.3} 2$, $c = \log_3 2$, 则 a, b, c 的大小关系是? (Given $a = \ln 3$, $b = \log_{0.3} 2$, and $c = \log_3 2$, the relationship among a, b, c is?)



- A. $a < c < b$
- B. $c < a < b$
- C. $a < b < c$
- D. $b < c < a$

42. 一个椭圆的中心在原点，长轴长是短轴长的 2 倍，且经过点 $P(-2,0)$ ，则该椭圆的方程是？ (An ellipse with center at the origin has a major axis that is twice the length of its minor axis and passes through point $P(-2,0)$. The equation of the ellipse is?)

- A. $\frac{x^2}{16} + \frac{y^2}{4} = 1$ 或 $\frac{x^2}{4} + y^2 = 1$ ($\frac{x^2}{16} + \frac{y^2}{4} = 1$ or $\frac{x^2}{4} + y^2 = 1$)
- B. $x^2 + \frac{y^2}{4} = 1$
- C. $\frac{x^2}{4} + \frac{y^2}{16} = 1$ 或 $\frac{x^2}{4} + y^2 = 1$ ($\frac{x^2}{4} + \frac{y^2}{16} = 1$ or $\frac{x^2}{4} + y^2 = 1$)
- D. $\frac{x^2}{16} + \frac{y^2}{4} = 1$

43. 已知向量 $\vec{a} = (-1, 2)$ 和 $\vec{b} = (3, -1)$ ，则 $3\vec{b} - \vec{a} = ?$ (Given vectors $\vec{a} = (-1, 2)$ and $\vec{b} = (3, -1)$, then $3\vec{b} - \vec{a} = ?$)

- A. $(0, -3)$
- B. $(0, 5)$
- C. $(-6, 5)$
- D. $(-6, 7)$

44. 若直线 l 与直线 $l_1: x + 2y + 4 = 0$ 垂直，且经过直线 $l_2: x + y + 1 = 0$ 和 $l_3: 2x + y - 1 = 0$ 的交点，则直线 l 的方程是？ (If line l is perpendicular to line $l_1: x + 2y + 4 = 0$ and passes through the intersection of lines $l_2: x + y + 1 = 0$ and $l_3: 2x + y - 1 = 0$, then the equation of l is?)

- A. $2x - y - 7 = 0$
- B. $x + 2y - 3 = 0$
- C. $2x + y - 5 = 0$
- D. $x - 2y - 6 = 0$

45. 已知复数 $|z| = 1$ ，则 $|2z + 1|$ 的最大值是？ (Given that the complex number $|z| = 1$, then the maximum value of $|2z + 1|$ is?)



1, the maximum value of $|2z + 1|$ is?)

- A. 2
- B. 1
- C. $\frac{5}{2}$
- D. 3

46. 对于等比数列 a_n , 其前 n 项和为 S_n , 若 $S_2 = 4$, $S_4 = 20$, 则 $S_8 = ?$ (For a geometric sequence a_n with sum of first n terms S_n , if $S_2 = 4$ and $S_4 = 20$, then $S_8 = ?$)

- A. 160
- B. 340
- C. 180
- D. 210

47. 有 6 个相同的球, 分别标有数字 1,2,3,4,5,6。无放回地随机抽取 3 个球, 每次抽取 1 个。

设 m 是前两个抽到的球上数字的平均数, n 是三个抽到的球上数字的平均数, 则 $|m - n| \leq \frac{1}{2}$ 的概率是? (There are 6 identical balls labeled with numbers 1,2,3,4,5,6. Three balls are randomly drawn without replacement, one at a time. Let m be the average of the numbers on the first two balls drawn, and n be the average of the numbers on all three balls drawn. The probability that $|m - n| \leq \frac{1}{2}$ is?)

- A. $\frac{13}{30}$
- B. $\frac{2}{3}$
- C. $\frac{8}{15}$
- D. $\frac{7}{15}$